INDEPENDENT CLOUD-READINESS REVIEW

Evidence before the estimate.

PreFlight is an offline, evidence-first review of a Guidewire-style policy administration estate. It inventories what exists, links claims to files, records every disposition, and names every unknown — before a migration estimate is signed.

OFFLINE · SYNTHETIC ESTATE · INDEPENDENT RESEARCH PROTOTYPE · NOT AFFILIATED WITH GUIDEWIRE

v1.0.0 · run R-DEMO-001 · demo-slice-300 · 2026-11-09

REVIEW PACKET
PKT-R-DEMO-001

6 SECTIONS

Files hashed and classified	300
Features traced	8
Evidence records	126
Advisory findings / rules	24 / 8
Explicit dispositions	12
Audit events	9 seed · 11 after 2 actions

THE PROBLEM

ESTIMATE-MOVING SCOPE

A migration estimate is a promise. The inventory is the proof.

The vendor estimate is not the risk by itself. The risk is the scope hidden underneath it: files that were never classified, features nobody traced, fields nobody disposed, and questions nobody owns.

> **Six to eight weeks** is a vendor estimate for APD product conversion — not a full migration estimate. It is a useful anchor. It is not an inventory of *your* estate.

01
What exists?
Every file, hash, parser status, and the four that cannot be parsed — listed, not dropped.

02
Which feature?
Cross-artefact links from PCF to entity, typelist, template, Gosu, and tests.

03
What is disposed?
An explicit source-to-target disposition, or a named gap. Never a silent rename.

04
What is unknown?
Owner, impact range, and the one question that would close the gap.

APD TYPES

24-character type limit → compile errors
APD limits **type names** to 24 characters. A longer name can be valid in source and still fail when endpoint generation compiles the generated code. The documented correction is a codegen `nameOverride` with the installed name.

Guidewire docs · “Correcting compile errors” · accessed 2026-11-09

PATTERNS

JDBC, explicit commits, servlets, hidden packages
These recur in migration lessons. They often surface in the CUE (the official Cloud Upgrade Estimate) or a build gate; the earlier they are found, the less the estimate moves.

Guidewire developer blog · migration lessons guest post · accessed 2026-11-09

SBT

Standards Based Template lines cannot convert
The Olos release note says SBT lines cannot be converted to APD-managed lines “at this time.” That is a readiness **scope and estimate risk**: it has to be priced or explicitly excluded, not assumed away.

Guidewire release highlights · Olos · accessed 2026-11-09

HANDOVER

Incomplete handover
Inventory lives in spreadsheets, heads, and ticket threads. Dispositions and assumptions are rarely written with an owner and a falsifier.

Field observation · not measured in this study

01
Scope becomes countable
300 files, 4 exclusions, 8 features, 126 evidence records.

02
Dispositions become explicit
12 source-to-target rows, 5 target changes, 0 _gen writes.

03
Unknowns become questions
Each gap gets an owner, an impact range, and a question to ask.

Source discipline: vendor claims are labelled as vendor claims; gated bodies are marked gated; “incomplete handover” is a design premise, not a measured result. Full list in `docs/presentation/sources.md`.

THE GAP

CAPABILITY MAP, NOT A BENCHMARK

No public tool we found combines PCF/entity/typelist/template/Gosu evidence into one disposition + assumption packet.

Guidewire CUE, Cloud Assurance Services, Lighthouse, SurePath, Developer Assistants, GoQu, CAST/ADDI for portfolio and code discovery, Syntra ETL for data migration and extraction, and SI assessments each cover a real slice. The gap is the hand-off: a file-level, cross-artefact feature inventory with a disposition ledger, an assumption register, and a reviewable packet.

CUE · CLOUD ASSURANCE SERVICES · LIGHTHOUSE · SUREPATH · DEVELOPER ASSISTANTS · GOQU · CAST / ADDI · SYNTRA ETL · SI ASSESSMENTS

TOOL / SERVICE	INVENTORY	FEATURE EVIDENCE	DISPOSITION LEDGER	ASSUMPTION REGISTER	REVIEW PACKET
Cloud Assurance Services official standards gate	code scan	no	no	no	findings
Guidewire CUE official Cloud Upgrade Estimate	scope	no	no	estimate risks	estimate
Lighthouse / SurePath / Assistants migration lab, delivery method, codegen help	lab	validated flows	no	no	handover
SonarQube / GoQu Gosu lint and quality	code only	no	no	no	issues
CAST / ADDI portfolio and code discovery	portfolio / code	dependencies	no	no	no
Syntra ETL data migration / extraction	data	no	no	no	no
Systems integrator services and expertise	bespoke	varies	varies	varies	varies
PreFlight offline, one-feature evidence review	300 files	cross-artefact	12 explicit	impact-rated	6 sections

THE ACTUAL GAP

The connective tissue, not another scanner.

Cross-artefact feature evidence + an explicit disposition ledger + an assumption register + a review packet. Each is adjacent to an existing tool; the integrated, reviewable chain is the missing artifact. Output: feature map · disposition ledger · assumption register · review packet.

DELIBERATELY NOT REPLACED

PreFlight does not compete with the official gate, the official estimate, the migration lab, or SI delivery.

CLOUD ASSURANCE · CUE · LIGHTHOUSE · SUREPATH · APD CONVERSION · CLOUD RETROFIT · SI DELIVERY

Positioning, not an independent product evaluation. Public product descriptions and documentation; see `sources.md` S-05, S-06, S-07, S-19–S-24.

THE ENGINEERED SOLUTION

FASTAPI · PYTHON 3.12 · JSON STORE

One pipeline. Nine stages. Every step leaves a trace.

A read-only snapshot is hashed, parsed, assembled into features, reviewed against cited rules, reconstructed into a local draft, compared, evaluated, and sealed into a review packet.

01 **Snapshot** — read-only tree; no live tenant

02 **Intake + hash** — manifest, status, before/after hash

03 **Parse** — PCF, entity, typelist, template, Gosu

04 **Discover** — features, evidence, gaps, confidence

05 **Review rules** — 8 rules, 24 hard negatives

06 **Reconstruct** — _ext draft, mapping, assumptions

07 **Compare** — preserved, changed, missing, unresolved

08 **Evaluate** — M1–M6, baselines, intervals

09 **Packet** — 6 sections, hashes, audit head

Read-only
Snapshot hashed before and after; source unchanged.

Unsupported listed
Binary, corrupt, and unsupported files are named.

Every claim evidence
Path, locator, excerpt, hash, method, confidence.

Every change pending
Draft changes wait for an explicit reviewer action.

No LLM required
Model off; deterministic retrieval; no external calls.

300 files → **126** evidence records → **24** findings / 8 rules → **12** dispositions → **1** review packet

one feature slice · one laptop · no telemetry

INPUT
Read-only snapshot
300 synthetic files; 4 excluded and named; source hash identical before and after.

OUTPUT
Evidence graph + packet
126 evidence records, 8 features, 24 cited findings, 12 dispositions, 1 packet.

BOUNDARY
Stops before deployment
No tenant, no LLM, no external calls, no _gen edit, no certification.

WHAT EACH STAGE WRITES

SNAPSHOT read-only source; no writes

INTAKE manifest · immutability · audit events

PARSE evidence records with locators

DISCOVER features · gaps · spine

REVIEW findings · review evidence

RECONSTRUCT mappings · changes · assumptions · draft

COMPARE comparison rows

EVALUATE metrics · intervals · baselines

PACKET packet JSON/HTML · audit head

DETERMINISTIC **Same snapshot + same rule pack → same evidence.** Reviewer decisions are the only mutation; every decision is hashed into the audit chain.

THE EVIDENCE MODEL

EVIDENCE BEFORE ASSERTION

A claim is only as strong as its locator, its label, and its open question.

PreFlight does not store “a finding.” It stores a claim with a source pointer, an excerpt hash, a method, a confidence label, and — where the evidence runs out — an assumption with an owner.

ANATOMY OF A CLAIM


```
{ "evidence_id": "EV-0001", "path":
"config/web/pcf/policy/PAVehicleDV.pcf", "locator": "xpath
/PCF/Screen/TypeKeyInput[1]:4", "excerpt": "<TypeKeyInput
id=\"VehicleType\" …/>", "excerpt_sha": "d76bf9908b1f…",
"method": "xml_xpath", "confidence": "parsed" }
```


WHY THE HASH MATTERS

The excerpt is not decorative. It is re-read from the snapshot and hashed; a fabricated or drifting excerpt fails the locator-support audit.

WHAT THE RECORD DOES NOT PROVE

- That the excerpt is semantically sufficient.
- That the parser understood every construct.
- That the decision is correct — only that it is inspectable.

CLAIM-STRENGTH LADDER

A label caps what may be inferred.

verified-id	12	Public standard ID/title verified; body may be gated.
verified-policy	3	Public policy or hardening guidance; no rule ID claimed.
verified-doc	3	Documented mechanism, e.g. the 24-character APD limit.
secondary	3	Credible secondary source, e.g. an experience report.
heuristic	3	PreFlight pattern; explicitly not a Cloud Standard.

Counts are the 24 advisory findings in R-DEMO-001. The separate SBT readiness flag is not one of the 24.

WHAT THE LABEL BUYS YOU

LABEL	YOU MAY SAY	YOU MAY NOT SAY
`verified-id`	This is standard ID/title X.	“This passes the gated rule body.”
`verified-policy`	Public policy discourages this.	“This is a Cloud Standards violation.”
`verified-doc`	The documented mechanism is X.	“The target estate behaves this way.”
`secondary`	An experience report flags this.	“This is an official rule.”
`heuristic`	PreFlight flags this pattern.	“Guidewire says this is wrong.”

ASSUMPTION REGISTER

ID	`A-01`
Statement	Territory_Ext exists; no rating or territory source artefact was found.
Owner	UNASSIGNED
Impact	high · heuristic range **3–5 engineer-days**
Question	Which rating table or service consumes Territory_Ext, and is it in scope?
Status	open

PACKET HASH / AUDIT CHAIN

`event 1` → `hash 1` → `event 2` → `hash 2` → `…` → `head`

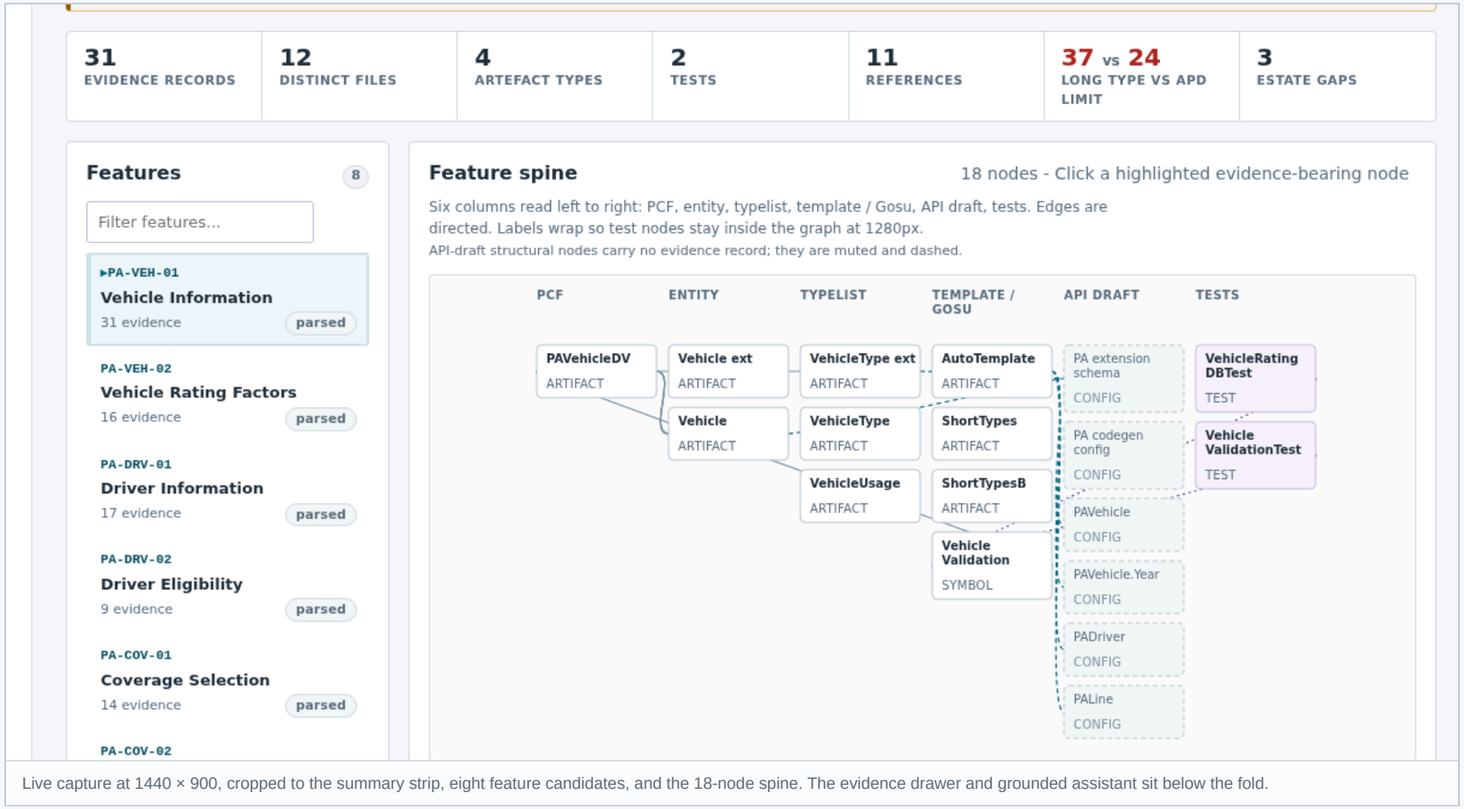
The packet hash covers the rendered packet HTML (hash fields normalized); `packet_json` separately covers canonical packet JSON; content digests bind findings, mappings, and evidence. The audit head covers the append-only event chain. Verification is local and tamper-evident — not non-repudiation.

REVIEWER QUESTION

If this evidence disappeared, which decision would become unsafe? If the answer is “none,” the record is not carrying its weight.

One feature, traced across six artefact families.

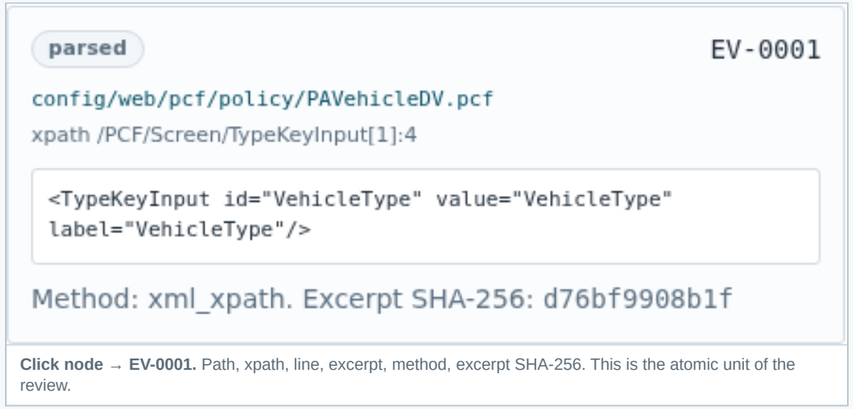
PA-VEH-01 Vehicle Information is the slice. The summary strip counts what the claim rests on; the spine shows how PCF, entity, typelist, template/Gosu, API draft, and tests connect.



Live capture at 1440 × 900, cropped to the summary strip, eight feature candidates, and the 18-node spine. The evidence drawer and grounded assistant sit below the fold.

1. **31 evidence / 12 files / 4 artefact types / 2 tests / 11 references — all PA-VEH-01 counts.** Every number is a count of linked records. The 37-character long type is separate: it appears in two files (template and product model) and is a PA-VEH-02 dependency, not one of the 11 references.
2. **Eight features, each with evidence count and confidence.** The selected feature is PA-VEH-01; the list is the inventory entry point.
3. **18-node spine, six columns.** PCF → entity → typelist → template/Gosu → API draft → tests. Directed edges; click any highlighted evidence-bearing node for evidence.
4. **37 vs 24, and 3 estate-wide gaps.** The long type is carried by two files and is a PA-VEH-02 dependency; the gaps are labelled estate-wide, not three missing links in this feature.

31	12	37:24	3
EVIDENCE	FILES	TYPE VS APD	ESTATE GAPS



Click node → EV-0001. Path, xpath, line, excerpt, method, excerpt SHA-256. This is the atomic unit of the review.

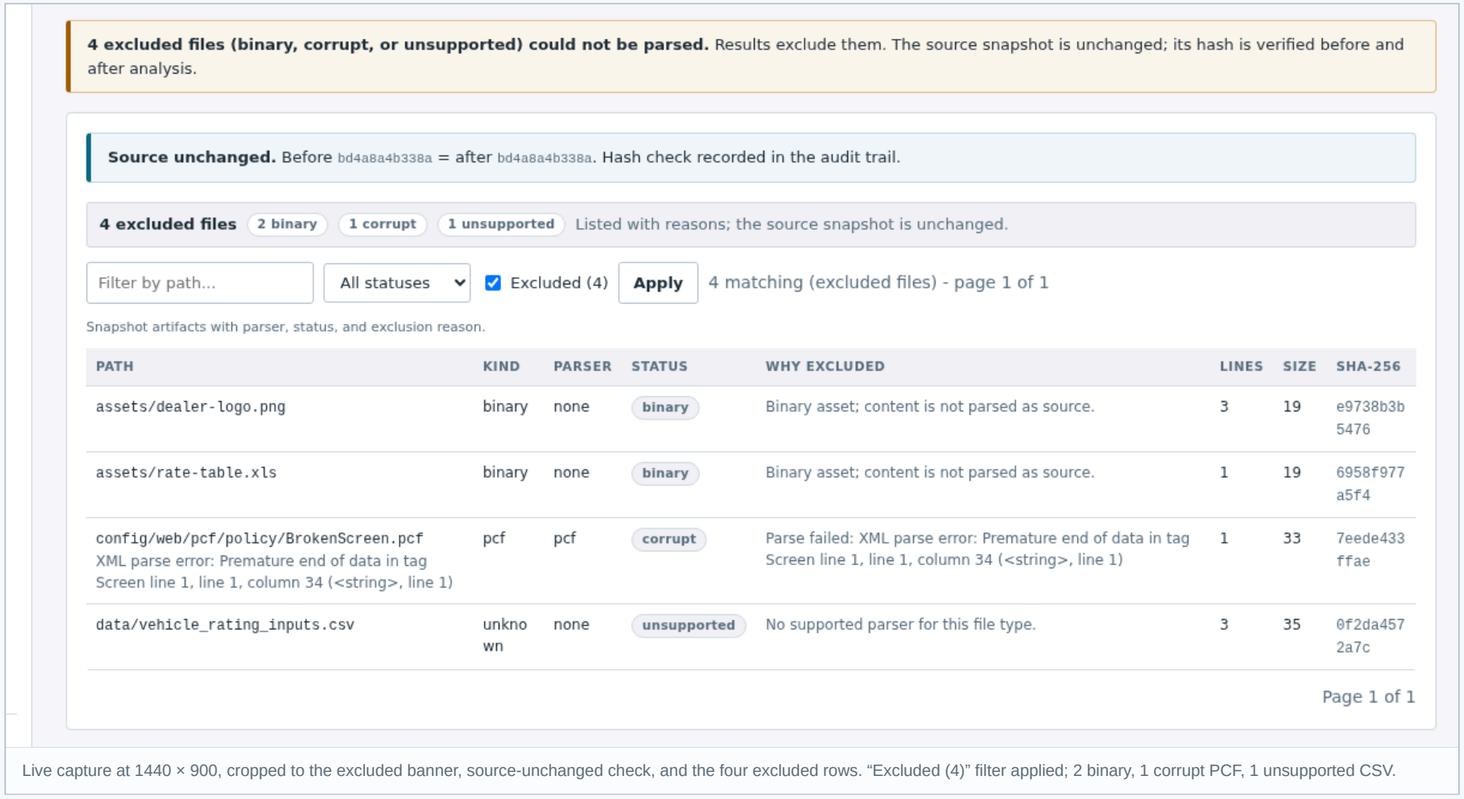
Fixture realism: the 300 files include generated filler. PCF, product model, and Gosu are synthetic sketches, not production Guidewire formats. Codegen and entity names match the fixture; this is not a claim of domain fidelity.

DEMO 02 · MANIFEST

No silent omission. Four files are excluded — and named.

A parser that quietly drops what it cannot read produces a confident lie. PreFlight records every file, status, reason, and hash; the source snapshot is checked before and after analysis.

300 FILES · 4 EXCLUDED



4 excluded files (binary, corrupt, or unsupported) could not be parsed. Results exclude them. The source snapshot is unchanged; its hash is verified before and after analysis.

Source unchanged. Before `bd4a8a4b338a` = after `bd4a8a4b338a`. Hash check recorded in the audit trail.

4 excluded files 2 binary · 1 corrupt · 1 unsupported — Listed with reasons; the source snapshot is unchanged.

Filter by path... | All statuses | Excluded (4) | Apply | 4 matching (excluded files) - page 1 of 1

Snapshot artifacts with parser, status, and exclusion reason.

PATH	KIND	PARSER	STATUS	WHY EXCLUDED	LINES	SIZE	SHA-256
assets/dealer-logo.png	binary	none	binary	Binary asset; content is not parsed as source.	3	19	e9738b3b5476
assets/rate-table.xls	binary	none	binary	Binary asset; content is not parsed as source.	1	19	6958f977a5f4
config/web/pcf/policy/BrokenScreen.pcf XML parse error: Premature end of data in tag Screen line 1, line 1, column 34 (<string>, line 1)	pcf	pcf	corrupt	Parse failed: XML parse error: Premature end of data in tag Screen line 1, line 1, column 34 (<string>, line 1)	1	33	7eede433ffae
data/vehicle_rating_inputs.csv	unknown	none	unsupported	No supported parser for this file type.	3	35	0f2da4572a7c

Page 1 of 1

Live capture at 1440 × 900, cropped to the excluded banner, source-unchanged check, and the four excluded rows. “Excluded (4)” filter applied; 2 binary, 1 corrupt PCF, 1 unsupported CSV.

1. **300 files; 4 excluded: 2 binary / 1 corrupt / 1 unsupported.** The manifest is the denominator for every later claim.
2. **Every excluded row has a reason and a SHA-256.** BrokenScreen.pcf fails XML parsing at line 1; `vehicle_rating_inputs.csv` has no supported parser. The filter is part of the UI, not a footnote.
3. **Before = after.** `bd4a8a4b338a` is identical before and after analysis; the immutable snapshot hash is also visible in the live banner. The check is recorded in the audit trail.

EXCLUDED FILE	STATUS	REASON
`assets/dealer-logo.png`	binary	not parsed as source
`assets/rate-table.xls`	binary	not parsed as source
`config/web/pcf/policy/BrokenScreen.pcf`	corrupt	XML parse error, line 1
`data/vehicle_rating_inputs.csv`	unsupported	no supported parser

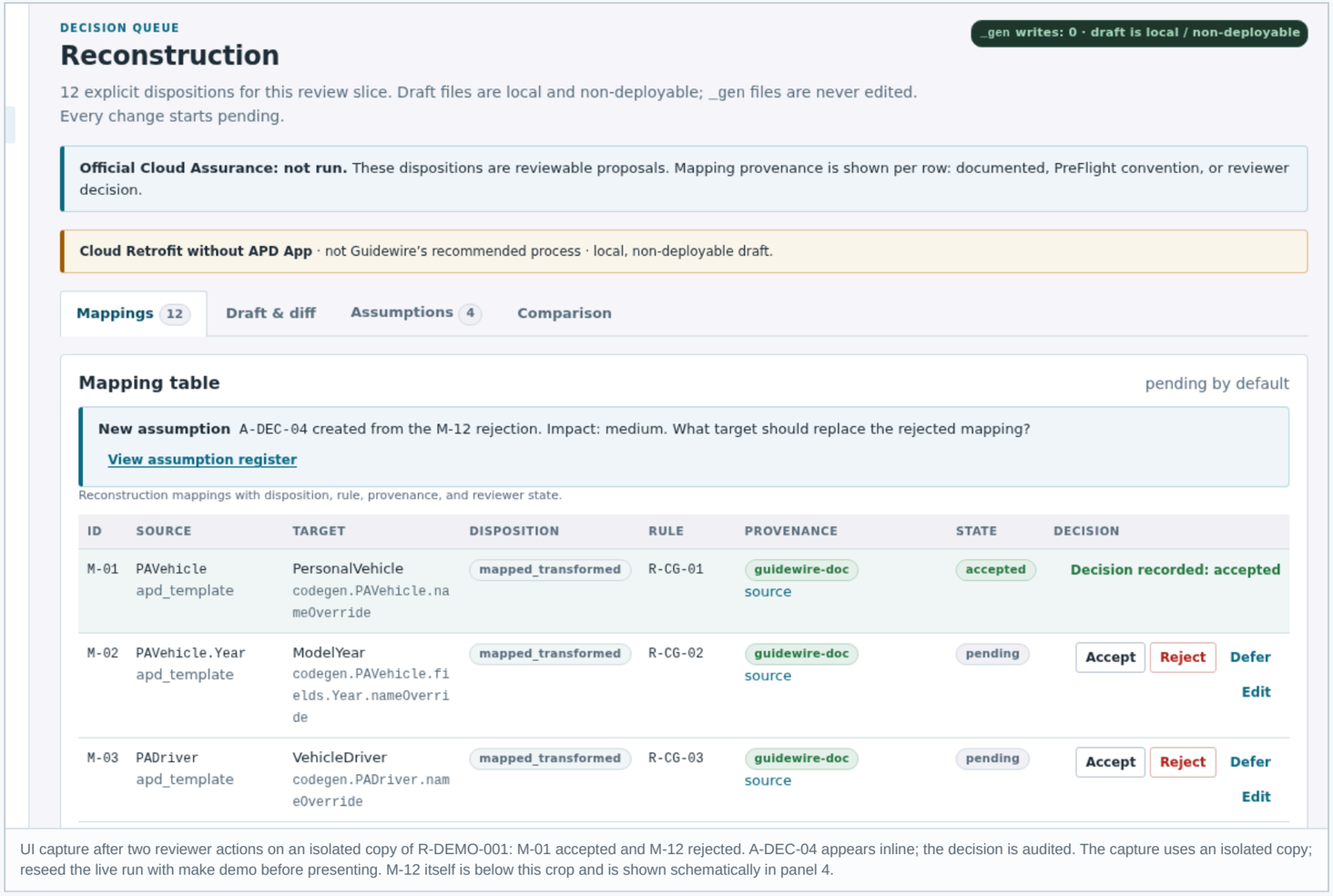
The discipline: an inventory without a failure list is a sample. The excluded files stay visible in the packet.

DEMO 03 · RECONSTRUCTION · AFTER REVIEWER ACTIONS

12 DISPOSITIONS · 0 _GEN WRITES

A decision queue, not a rename script.

Twelve explicit source-to-target dispositions. Provenance is per row: documented example, PreFlight convention, or reviewer decision. Reject a row and the unresolved question becomes a named assumption.



UI capture after two reviewer actions on an isolated copy of R-DEMO-001: M-01 accepted and M-12 rejected. A-DEC-04 appears inline; the decision is audited. The capture uses an isolated copy; reseed the live run with make demo before presenting. M-12 itself is below this crop and is shown schematically in panel 4.

1. **Reject M-12 → new A-DEC-04.** "What target should replace the rejected mapping?" Impact medium, owner unassigned, status open.
2. **Accept is explicit and recorded in the audit.** M-01 row turns green only after the action; the other rows stay pending.
3. **_gen writes: 0.** Draft is local and non-deployable; only `_ext`, codegen, and product-access proposals are written.
4. **M-12 rejected, not deleted.** 12 dispositions / 5 target changes. Four exact documented examples + one documented 24-char mechanism on a synthetic long type; six API conventions; one typelist convention.

12	5	4	0
DISPOSITIONS	TARGET CHANGES	ASSUMPTIONS AFTER REJECT	_GEN WRITES

SCHEMATIC · M-12 ROW BELOW THE CROP

ID	`M-12` · rejected
Source	`VehicleType.Motorcycle_Ext`
Target	`schema.vehicleType.motorcycle`
State	Stays in the ledger; A-DEC-04 carries the open question.

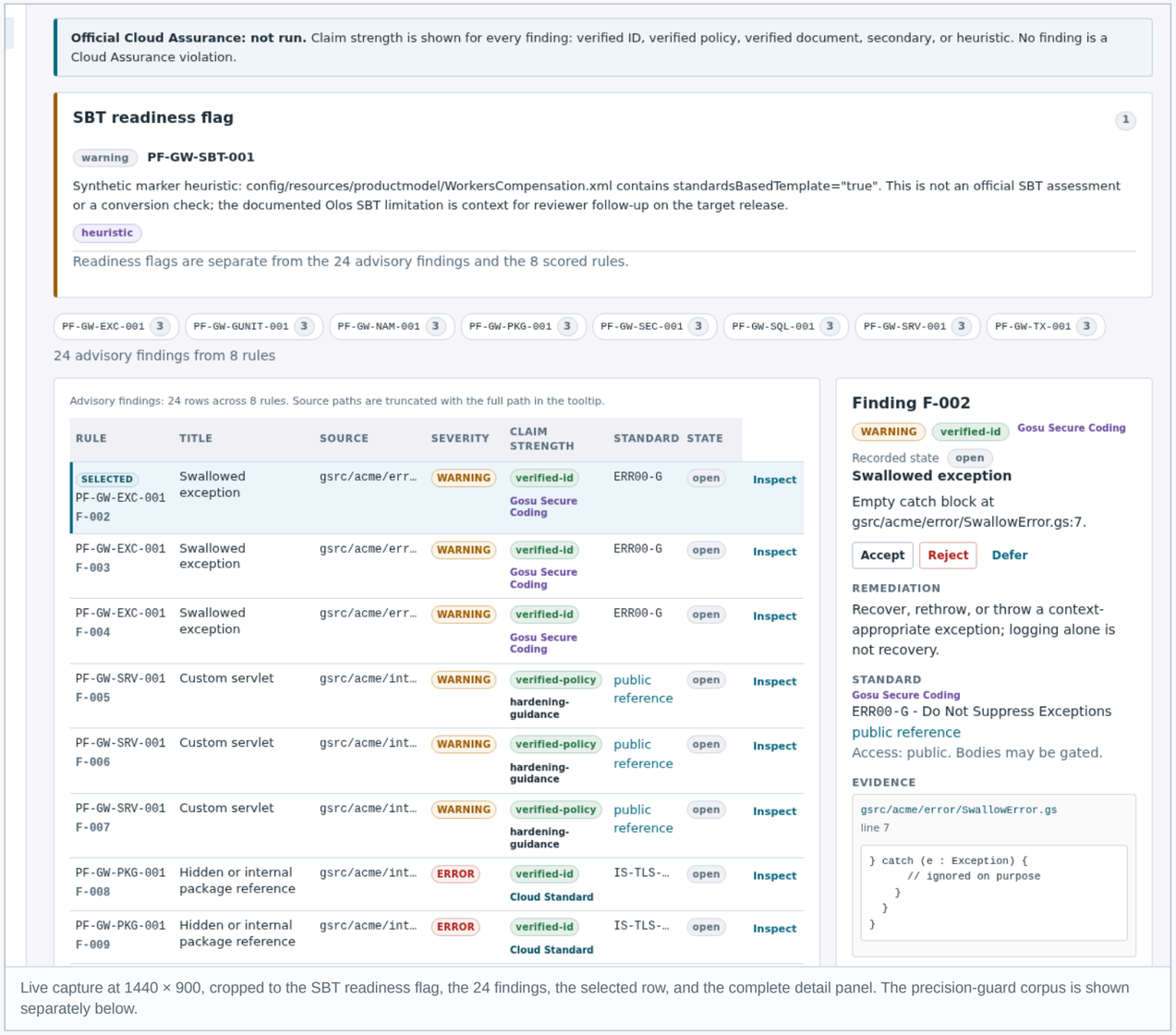
Scope: draft targets Cloud Retrofit **without APD App**, which Guidewire explicitly labels not the recommended process; field-level `nameOverride` applies only on that path. The `_ext` draft is a local JSON contract, not official Guidewire schema, mapping, or updater syntax.

DEMO 04 · FINDINGS

OFFICIAL CLOUD ASSURANCE: NOT RUN

Precision is a feature. Restraint is the interface.

Twenty-four advisory findings from eight rules. Every row carries a claim-strength label. The rule pack ships 24 hard negatives that must stay silent — including the supported `Query.make` API.



Live capture at 1440 × 900, cropped to the SBT readiness flag, the 24 findings, the selected row, and the complete detail panel. The precision-guard corpus is shown separately below.

1. **24 findings / 8 rules.** 3 per rule; the guard corpus is separate: 8 rules × 3 hard negatives = 24 cases.
2. **Claim strength on every row.** 12 verified-id, 3 verified-policy, 3 verified-doc, 3 secondary, 3 heuristic.
3. **Two standard families, not one.** Cloud Standards: `IS-TLS-1238` (SQL/JDBC), `IS-TLS-1191` (hidden package). Gosu Secure Coding: `SEC00-G` (secrets), `ERR00-G` (swallowed exception). The `standard_family` field in the data distinguishes `cloud-standard` from `gosu-secure-coding`.
4. **SBT is a separate readiness flag.** Not mixed into the 24 advisory findings or the 8 scored rules.
5. **24 hard negatives stay silent.** `Query.make(PersonalVehicle).select()` is supported access, not JDBC. Comments and strings are masked first.

PF-GW-SQL-001

Direct JDBC or SQL access

- `Query.make(PersonalVehicle).select()`
  supported data-access API, not JDBC
- `DriverManager.getConnection(...)` in a comment or string
  comments and strings are masked before matching
- `client.executeQuery("search")`
  non-JDBC search call is not direct SQL access

Precision guard. 8 rules × 3 near-misses; `Query.make` is supported access, not JDBC.

No violation claim. Advisory, pattern-based, cited. Not Cloud Assurance, a pass/fail, or legal/security advice.

DEMO 05 · GROUNDED ASSISTANT

It would rather refuse than invent a source.

RETRIEVAL-ONLY · MODEL OFF

Deterministic retrieval over the run's evidence. Model off. A missing rating table produces a refusal with the missing sources named; a present typelist entry produces a citation card with path, locator, and excerpt.

REFUSAL **"Does this estate use territory rating?"**

You: Does this estate use territory rating?

> Cannot determine - no rating table or territory source artefact exists in this snapshot. Searched scopes: product model, rating Gosu, rate tables, entity extensions. No evidence in this snapshot supports a different answer.
>
> Would need: a rating table or territory factor source; rating input export; the service that consumes Territory_Ext

1. **Names the missing evidence, not a guess.** "Would need: a rating table or territory factor source; rating input export; the service that consumes Territory_Ext."
2. **Territory is A-01 in the packet.** The refusal and the assumption register point at the same unresolved source.

CITED ANSWER **"Which files reference Motorcycle_Ext?"**

You: Which files reference Motorcycle_Ext?

Motorcycle_Ext appears in 1 evidence record, including the typelist extension. The target enum disposition is a reviewer decision, not a source fact. Each citation resolves to a file and line/pointer.

Citations (open the evidence panel)

linked EV-0023 **Open evidence →**

`config/extensions/typelist/VehicleType.ttx`

xpath /typelistExtension/code[1]:3

```
<code name="Motorcycle_Ext"/>
```

3. **A citation card is an evidence record.** `EV-0023` → `config/extensions/typelist/VehicleType.ttx` → xpath `/typelistExtension/code[1]` → line 3.
4. **Disposition is still a reviewer decision.** The assistant reports the source fact; it does not accept the mapping.

No generation
No LLM, no tools, no external calls. Retrieval over evidence IDs.

No orphan citations
Every cited ID resolves to the embedded evidence record, or is labelled missing.

No authority transfer
An answer is evidence to review, not a decision. The human still accepts or rejects.

DEMO 06 · EVALUATION

NOT A BENCHMARK

Perfect scores measure the instrument. The estate is still unproven.

Six headline metrics on the in-sample demo slice, with raw counts and 95% Wilson intervals. The holdout is pre-computed and never imported. A small n with a 100% point estimate is not certainty.

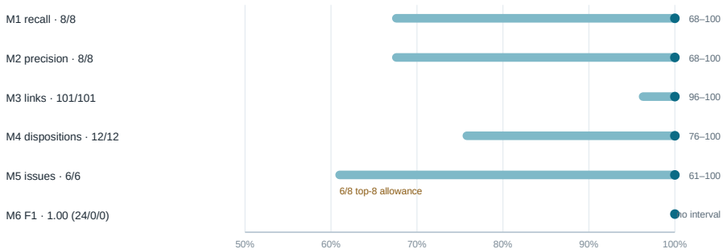
8/8 M1 RECALL	**8/8** M2 PRECISION	**101/101** M3 LINKS + 30/30 SAMPLE
12/12 M4 DISPOSITIONS	**6/6** M5 ISSUES · 6/8 SLOTS	**1.00** M6 F1 · 24 TP / 0 FP / 0 FN

Guard count, not a seventh metric: 24/24 hard negatives silent.

Holdout: 4/4 feature recall, 95% Wilson 51.0%–100.0%, four-feature naming-shift fixture. **Not imported** into the demo or used for tuning.

95% Wilson intervals

dot = observed value · line = interval · M6 has no interval (bootstrap deferred)



Wilson score intervals from reported numerators and denominators. M6 has no interval at this n; bootstrap deferred. Demo metrics are in-sample on one synthetic fixture. 24/24 hard negatives are a separate guard count.

PRE-REGISTERED MARGINS VS BASELINES

METRIC	BASELINE	DELTA	REQUIRED	STATUS
M1 feature recall	PCF + grep: 87.5%	+12.5 pp	+10 pp	PASS
M4 dispositions	Identity mapper: 50.0%	+50.0 pp	+10 pp	PASS
M6 finding micro-F1	Grep-only F1: 84.0%	+16.0 pp	+10 pp	PASS

WHAT THE NUMBERS DO AND DO NOT SAY

- \+ **M3 is 101/101 all links** plus a 30/30 stratified sample. It checks excerpt support at the source, not semantic entailment.
- \+ **M5 is 6/6** but the allowance is **6 of 8 slots**. It is a fixed two-decoy top-8, not a learned ranking.
- \+ **M6 is F1 1.00 (24 TP / 0 FP / 0 FN)** = 48/48 micro-F1. Line-level matching is deferred; the demo uses path-level matching.
- – **No human IAA.** The reference target is internally authored and AI-assisted, not blind and independent.
- – **One synthetic fixture.** The four-feature holdout is naming-shift only and never imported or tuned.
- – **Not a benchmark.** No real estate, customer code, deployment, or official Cloud Assurance run.

WHAT WOULD CHANGE THE VERDICT

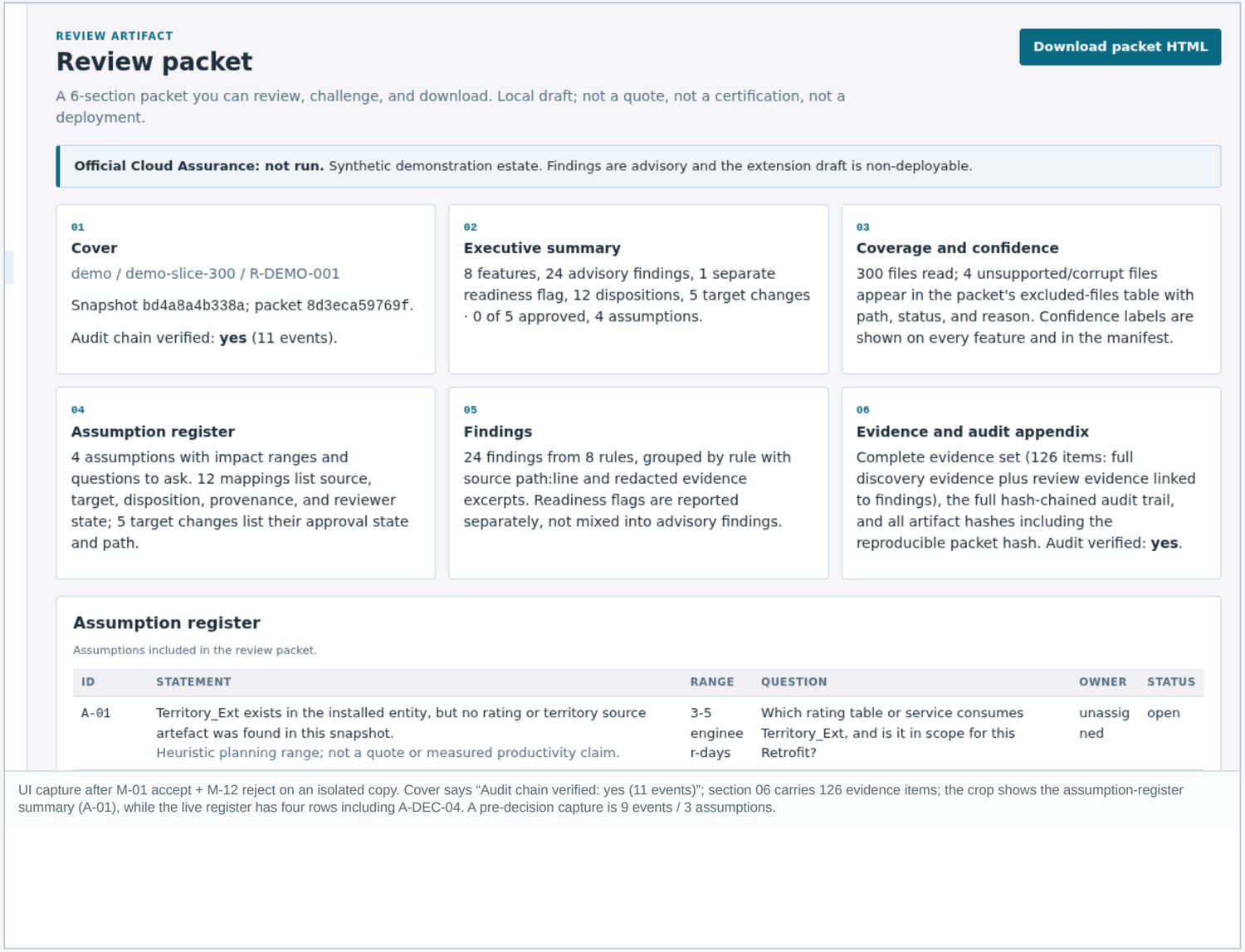
- A real anonymised estate and a second LOB.
- Blind human annotation with reported IAA.
- Line-level finding evaluation and compile validation.

DEMO 07 · REVIEW PACKET · AFTER REVIEWER ACTIONS

126 EVIDENCE · 11 AUDIT EVENTS

The review artifact is the product.

Six sections, every excluded file, every assumption, every disposition, 126 evidence items, and a hash-chained audit trail. This capture is post-decision: 11 audit events and 4 assumptions after Accept M-01 + Reject M-12. Downloadable HTML; locally verifiable; explicitly not a certification.



UI capture after M-01 accept + M-12 reject on an isolated copy. Cover says “Audit chain verified: yes (11 events)”; section 06 carries 126 evidence items; the crop shows the assumption-register summary (A-01), while the live register has four rows including A-DEC-04. A pre-decision capture is 9 events / 3 assumptions.

1. **Downloadable HTML.** A local, self-contained review artifact — not a dashboard link that expires.
2. **Six sections:** cover · executive summary · coverage and confidence · assumption register · findings (24 + 1 separate readiness flag) · evidence and audit appendix.
3. **126 evidence items** = full discovery evidence plus review evidence linked to findings. Not a 20-row sample.
4. **Audit verified: yes, 11 events post-decision.** Seed state is 9 events; Accept M-01 + Reject M-12 add two. The packet hash seals the rendered packet HTML; `packet_json` seals canonical JSON; content digests bind findings, mappings, and evidence; the audit head seals the event chain. Two-step verification: `verify-packet` checks the packet and content digests; `verify-audit` checks the audit chain only, not the packet. Local tamper-evident, not non-repudiation.

01 Cover + hashes	02 Executive summary
03 Coverage & confidence	04 Assumption register
05 Findings	06 Evidence & audit

12 mappings · 4 assumptions · 1 readiness flag · 4 excluded files · 11 audit events

```
packet      read live rendered packet HTML
packet_json read live canonical JSON
audit head  read live event-chain head
snapshot    bd4a8a4b338a… fixture; stable
```

```
$ preflight verify-packet R-DEMO-001
```

```
$ preflight verify-audit R-DEMO-001
```

Content digests bind findings, mappings, and evidence. Packet, packet_json, and audit-head values change with reviewer actions and packet refreshes — read them from the live screen. `verify-packet` checks the packet; `verify-audit` checks the audit chain only, not the packet. The bundled packet is a capture-time example, not a live artifact.

ENGINEERING RIGOR

INTERNAL ADVERSARIAL REVIEW

The demo is the surface. The hardening is the work.

416 tests and a Playwright golden path. **18/18** recorded freeze hashes recompute. A security contract with archive, XML, YAML, CSRF, CSP, and redaction tests. Adversarial review found real flaws; each row below was fixed and regression-tested.

416 tests collected | **1** Playwright golden path | **18/18** frozen hashes recompute | **0** raw secret values in artifacts | **9** seed audit events verified

FLAW FOUND	BEFORE	AFTER / CONTROL
Concurrent mapping decisions	Simultaneous accepts could lose writes	12/12 persisted; per-run lock + transaction; 409 on conflict
Secret leakage through evidence	Parser warnings / adjacent context could store raw values	Redact at capture; fingerprint only; packet and logs tested
YAML alias bomb	Recursive aliases could expand without bound	Aliases rejected before expansion; depth/node caps; sub-second test
Forged evidence scoring	Fabricated excerpts could still match gold paths	48/48 → 0/48 when excerpts are forged; locator support required
Packet evidence completeness	A 20-row discovery sample	126 complete evidence items + audit appendix
Evidence audit scope	A 30-link stratified sample was the headline	101/101 all links + 30/30 sample; fabricated excerpts fail
Audit verification cost	Repeated chain scans and no truncation checkpoint	Single-pass walk + checkpoint head; 9 seed events verified
Mobile layout	Horizontal overflow at 390 px	Clipped overflow + responsive regression coverage

127.0.0.1 BIND | HOST / ORIGIN ALLOWLIST | CSRF SYNCHRONISER TOKEN | CSP SELF-ONLY | ZIP-SLIP / BOMB REJECTION | XML ENTITIES OFF | YAML TAGS REJECTED | HASH-CHAINED AUDIT | NO TELEMETRY

NOVELTY & PRIOR ART

NO “FIRST / ONLY” CLAIM

The algorithms are not new. The integration and the evidence discipline are the contribution.

Feature location, traceability recovery, architecture conformance, and assurance cases are mature fields. Reversa already combines traceability, confidence marking, gap preservation, and a SHA-256 manifest. PreFlight’s difference is domain-specific integration, not a new algorithm.

CLOSEST PRIOR ART

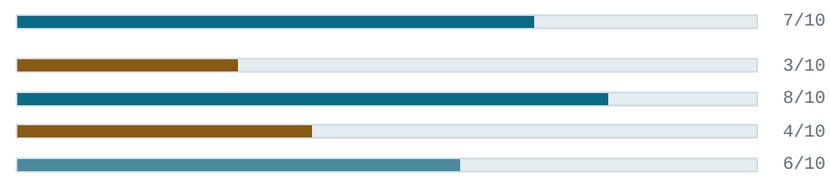
AREA	REPRESENTATIVE WORK	WHAT IT ESTABLISHES
Feature location	Dit et al. 2013; Marcus et al. 2004	IR and concept location can recover where a feature lives.
Traceability	Antoniol et al. 2002; FTLR 2021; LiSSA 2025; TraceLLM 2026	Links between artefacts can be recovered, ranked, and evaluated.
Reverse documentation	Reversa, arXiv:2605.18684 (2026)	Traceable specs, confidence marking, gaps for human validation, manifest.
Architecture conformance	Murphy et al. reflexion models (1995)	Compare intended and observed structure; surface drift.
Assurance cases	Kelly & Weaver, GSN (2004)	Claims must be supported by argument and evidence.
Tooling	GoQu, CAST, ADDI, Guidewire CUE	Gosu lint, portfolio discovery, official estimate.

Selective, not exhaustive. Public IDs and titles only; gated bodies marked gated. Vendor claims labelled as vendor claims. We do not claim the absence of prior work. Citations in `docs/presentation/sources.md`. Reversa is the closest single work; PreFlight is narrower and domain-specific.

Threat to the novelty claim: if Reversa adds Guidewire artefact adapters and a disposition ledger, the gap narrows quickly. The moat is not the algorithm. It is the domain evidence model, the review workflow, and the offline independence.

HONEST SCORECARD

Author’s assessment, not a benchmark or external review.



Research novelty is deliberately low: the individual mechanisms are established. The contribution is the integration, the Guidewire artefact model, and the review artifact.

WHAT IS ACTUALLY DIFFERENT

1. Cross-artefact feature evidence for Guidewire-style PCF/entity/typelist/template/Gosu.
2. A disposition ledger with per-row provenance and reviewer state.
3. An impact-rated assumption register tied to gaps.
4. A self-verifying review packet, offline, before the official estimate.

VIABILITY

RESEARCH PROTOTYPE

The wedge is real. The evidence is not yet sufficient to call it a product.

A research prototype today. The defensible path is an independent, offline, pre-CUE evidence packet for one LOB, used as a systems-integrator accelerator — not a replacement for the official estimate or gate.

KILL RISKS / OPEN EVIDENCE GAPS

- **Synthetic in-sample**
  One 300-file fixture; not a real estate.
- **Internally authored target**
  AI-assisted reference, not blind independent annotation.
- **No human IAA**
  AI-AI concordance would not be human inter-annotator agreement.
- **No real estate**
  Customer code, scale, and naming drift unproven.
- **CUE is official**
  Official Cloud Upgrade Estimate; PreFlight prepares file-level evidence before it.
- **4-feature holdout**
  Naming-shift only; never imported or tuned.
- **Vendor services and assistants**
  Cloud Assurance Services, Lighthouse, and Developer Assistants already own parts of the workflow.
- **Gated rule corpus**
  Public IDs/titles; bodies and threshold are gated.
- **Parser / naming shift**
  Custom PCF shapes and renames can break matching.
- **Draft non-deployable**
  Local _ext draft is not compiled or tenant-validated.

PROPOSED ACCEPTANCE TESTS BEFORE THE NEXT CLAIM

TEST	QUESTION	PASS SIGNAL
Desirability	Would an SI reuse the packet on another LOB?	Requested, not subsidised
Feasibility	Do supported parsers cover a real LOB?	≥ 90% of files parsed
Credibility	Do blind humans agree with the links?	IAA κ ≥ 0.6 on 30 links

WHAT WE WOULD NEED TO BELIEVE

- An SI will pay for independent pre-estimate evidence, not just delivery hours.
- A real estate produces findings and dispositions comparable to the synthetic slice.
- Reviewers prefer the packet to the spreadsheet they use today.

DEFENSIBLE WEDGE

Vendor-independent, offline, pre-CUE, file-level evidence for one LOB.

Why it can survive: independence from the vendor gate, auditability, low deployment friction, and reviewer control. “Independent” means vendor-independent, **not independently validated**: the evaluation is internal and in-sample. The buyer is an SI or an insurer’s migration lead who needs a repeatable evidence pack before CUE and Cloud Assurance Services fix the estimate and the gate.

VERDICT

Research prototype today. Viable as a wedge if a pilot estate and blind human annotation arrive. Not viable as a standalone product on the current evidence.

ROADMAP TO A CREDIBLE CLAIM

01 Real anonymised estate, one LOB

02 Second LOB to test transfer

03 Blind human annotation + IAA

04 Line-level findings

05 Performance envelope at estate scale

06 Cloud Assurance / CUE import-export

07 Compile validation of draft

CLOSE

Every claim has evidence.
Every unknown has a name.
Every change waits for a human.

That is the standard PreFlight is trying to make ordinary: an estimate you can inspect, challenge, and sign — with the unknowns written down instead of discovered at week five.

Pilot estate
An anonymised, authorised snapshot of one LOB. The next experiment is a real inventory, not another synthetic fixture.

SME annotation
Blind review by two domain experts, with disagreements adjudicated. That is the human IAA the current evaluation lacks.

Feedback
Attack the claims, the packet, and the disposition model. The fastest way to improve this is to try to break it.

Pilot ask: one authorised estate, two blind SME annotators, hard feedback.

Saundarya Singh · Independent research prototype **Repository** github.com/SaundaryaSingh/Preflight-CloudMigration **License** MIT **Deck** docs/presentation/index.html

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